Ujjwal Sood

Arlington, TX | (216) 418-5404 | ujjwalsood96@gmail.com | LinkedIn | GitHub | Portfolio

Professional Profile

Full-stack and AI/ML engineer shipping production web platforms, voice-enabled LLM systems, computer-vision tools, and browser-native media pipelines with React, TypeScript, FastAPI, Python, MongoDB, Docker, QLoRA, and vLLM.

Education

University of Texas at Arlington — **Arlington, TX**
Bachelor of Science in Computer Science — *January 2023 - December 2026*

Experience

UTA Parking & Transportation Services — **Arlington, TX**
Full-Stack Developer — *May 2026 - Present*

- Accomplished **98.3% faster event-parking planning** as measured by reducing PCO processing from **120 minutes to 2 minutes**, by engineering an internal planner with interactive maps, accessibility states, CSV/ICS imports, Outlook exports, and automated guest packets and emails.
- Accomplished **campus-wide parking discovery** as measured by **60 mapped lots across 6 eligibility modes**, by building a live permit navigator with arrival-time filtering, destination search, shuttle support, and normal or satellite map views.
- Accomplished **occupancy-informed parking decisions** as measured by **3 live factors: permit, location, and occupancy**, by integrating Mistall sensor and camera data with configurable lot mappings, stale-cache fallback, and interactive permit layers.

Projects

FinalRound AI Interviewer — Live App | GitHub
Next.js, TypeScript, FastAPI, MongoDB, WebSockets, Qwen2.5, QLoRA, vLLM, Tauri, Rust, Docker

- Accomplished **personalized voice-interview simulation** as measured by **9 mode-difficulty combinations across 10 application routes**, by building resume- and job-description-aware workflows with Next.js, FastAPI, WebSockets, Groq Whisper, and Kokoro TTS.
- Accomplished **40.6% stronger model convergence** as measured by reducing evaluation loss from **2.01 to 1.194**, by constructing 300K+ dialogue tokens from 47 interview transcripts, fine-tuning Qwen2.5-7B with QLoRA, and serving the adapter through vLLM.
- Accomplished **automated coding assessment** as measured by a **3,957-problem bank with test grading and saved attempts**, by integrating Monaco Editor, MongoDB caching, and sandboxed Piston execution.

Personal Finance Manager — GitHub
React 19, TypeScript, Vite, FastAPI, SQLAlchemy, SQLite, JWT, Pytest

- Accomplished **end-to-end personal-finance management** as measured by **31 REST endpoints across 8 product domains**, by engineering JWT-secured FastAPI services for accounts, transactions, budgets, goals, recurring payments, reports, alerts, and affordability analysis.
- Accomplished **complete responsive financial workflows** as measured by **10 React pages, 20 reusable components, and 9 typed API clients**, by building dashboards, transaction entry, budget tracking, savings progress, reports, authentication, and account settings in React 19 and TypeScript.
- Accomplished **verified backend reliability** as measured by **11 of 11 passing Pytest tests**, by testing registration, budgets, savings goals, monthly reports, and affordability calculations against isolated SQLite state.

AI Blood Test Analyzer — GitHub
React, FastAPI, MongoDB, Gemini, JWT, PDF Processing, Docker, Pytest

- Accomplished **resilient biomarker interpretation** as measured by **40 built-in reference ranges and 2 analysis paths**, by combining Gemini multimodal PDF analysis with a local rule-based fallback that flags high, low, and normal results.
- Accomplished **secure report processing and persistence** as measured by **11 API routes and 8 of 8 passing tests**, by implementing access/refresh JWTs, MongoDB history, PDF uploads, automatic token refresh, and Pytest coverage.

Reel Maker — Live Demo | GitHub
JavaScript, WebAssembly, ffmpeg.wasm, Web Audio API, Canvas, Three.js, Firebase

- Accomplished **private, serverless highlight generation** as measured by support for **500 MB or 5-minute videos and 15-60 second reels**, by executing decode, trim, crossfade, stitch, preview, and download operations entirely in-browser with ffmpeg.wasm.
- Accomplished **automated moment selection** as measured by **3 weighted signals: audio, motion, and scene change**, by building Web Audio and Canvas analyzers with tunable scoring and non-overlapping clip selection.

Hand Gesture Control — GitHub
Python, OpenCV, MediaPipe, PyAutoGUI

- Accomplished **touchless desktop interaction** as measured by mapping **21 MediaPipe hand landmarks into 2 verified OS actions**, by translating webcam coordinates into PyAutoGUI cursor movement and pinch-click gestures with confidence thresholds, cooldown control, and a fail-safe.

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, C, C++, SQL
Frontend & Backend: React, Next.js, Vite, Tailwind CSS, FastAPI, Express, Node.js, REST APIs, WebSockets, JWT
AI/ML: Qwen, QLoRA, PEFT/LoRA, vLLM, Gemini API, OpenAI API, NLP, scikit-learn, Computer Vision
AI Developer Tools: Claude Code, Codex, Gemini CLI, Antigravity
Cloud & Data: MongoDB, PostgreSQL, MySQL, SQLite, SQLAlchemy, Docker, Firebase, Vercel, AWS, Azure, Linux
Engineering: Git, GitHub Actions, CI/CD, Pytest, Agile/Scrum, System Design, OOP, Figma, Jira

Certifications

Microsoft Python Programming Fundamentals | **IBM** Artificial Intelligence Essentials V2 | **IBM** Introduction to Mobile App Development

Leadership

Students in Computing and Artificial Intelligence, UT Arlington — *Member*
Accomplished broader AI/ML engagement as measured by project demonstrations to **50+ students and faculty**, by presenting technical work, helping organize seminars, and supporting peer learning in emerging software and AI tools.